CYC Standard Data Format

What is a CYC file?

A CYC (Cycle Counting) file contains the counted fatigue cycles obtained from an LDS file using a selected cycle counting method.

File naming convention

File: CYC_{ResearcherLastname}_{YYYY-MM}_{method}.csv

- YYYY-MM = starting month of the experiment
- method = range_mean, range_pair, simplified_rainflow, etc.
- Example: CYC_Dupont_2024-03_range_mean.csv

File format requirements

- Encoding: UTF-8
- Separator: comma (,)
- File type: .csv
- Header required
- Units: MPa

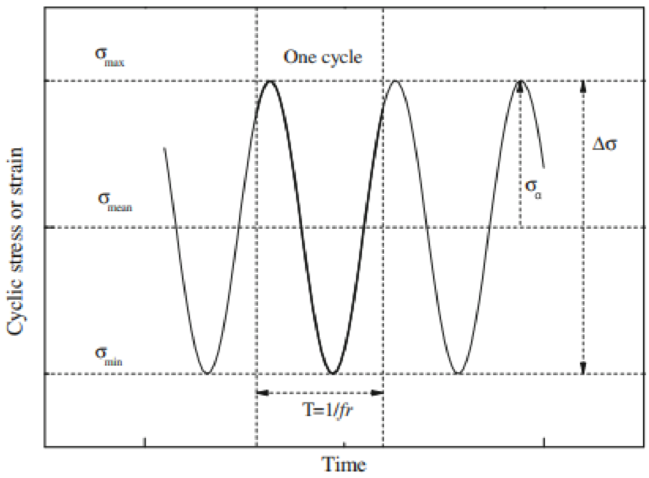


Figure 1 : Basic fatigue terminology (Vassilopoulos & Keller, 2017).

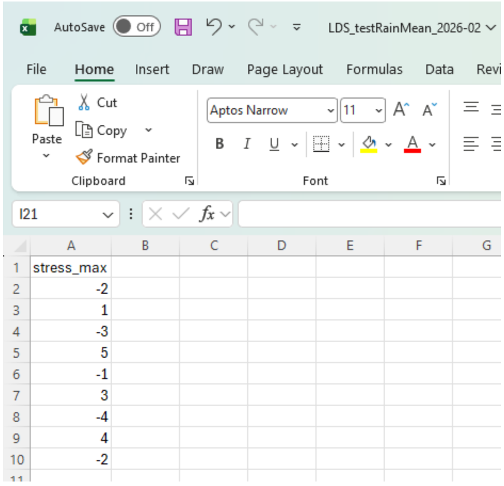
$$R = \frac{\sigma_{min}}{\sigma_{max}}$$

$$R = \frac{\sigma_m - \sigma_a}{\sigma_m + \sigma_a}$$

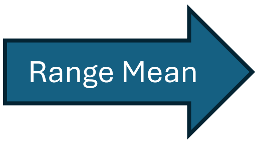
Required columns (column names must be exact)

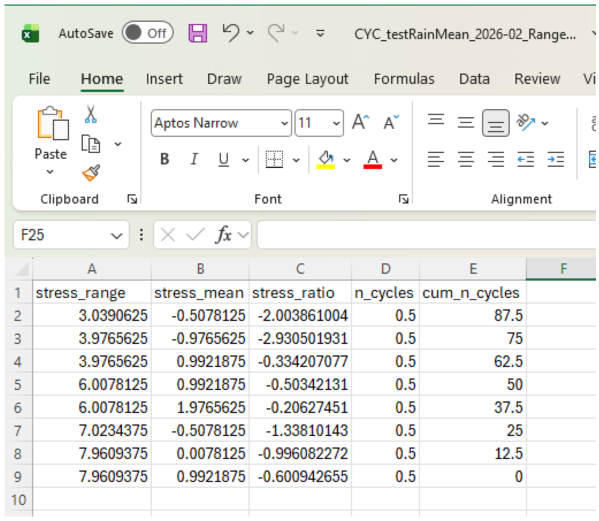
Variable name	Description	Symbol	Unit	Data type	Mandatory
stress_range	Stress range (= 2 x σ_a)	$\Delta\sigma$	[MPa]	double	y
stress_mean	Mean stress	σ_m	[MPa]	double	y
stress_ratio	Stress ratio	R	[-]	double	y
n_cycles	Number of cycles	n		double	y
cum_n_cycles	cumulative number of cycles for a given bin of σ_a, σ_m and R.			double	y

Example of valid CYC file



A
stress_max
-2
1
-3
5
-1
3
-4
4
-2





stress_range	stress_mean	stress_ratio	n_cycles	cum_n_cycles
3.0390625	-0.5078125	-2.003861004	0.5	87.5
3.9765625	-0.9765625	-2.930501931	0.5	75
3.9765625	0.9921875	-0.334207077	0.5	62.5
6.0078125	0.9921875	-0.50342131	0.5	50
6.0078125	1.9765625	-0.20627451	0.5	37.5
7.0234375	-0.5078125	-1.33810143	0.5	25
7.9609375	0.0078125	-0.996082272	0.5	12.5
7.9609375	0.9921875	-0.600942655	0.5	0

LDS file **CYC file**

For the CSV file, Microsoft Excel works well for editing and saving the data.
However, other tools such as LibreOffice Calc, Google Sheets, or any text editor (e.g., Notepad++, VS Code) can also be used, provided the file is saved in CSV format (UTF-8 encoding).